

PRUDHOE BAY HARBOR, AK, USA

WMO: 997386

Lat: 70.400N

Lon: 148.527W

Elev: 16

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.6	(c) -36.8	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 50.6	(k) -1.6	(l) 46.7	(m) -1.8	(n) 9.1	(o) 270	(p) 1.169

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.7	(c) 56.8	(d) N/A	(e) 53.2	(f) N/A	(g) 49.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 11.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 43.3	(b) 38.8	(c) 34.2	(d) DB	(e) -42.1	(f) 64.6	(g) 3.7	(h) 4.1	(i) -44.7	(j) 67.5	(k) -46.8	(l) 70.0	(m) -48.9	(n) 72.3	(o) -51.6	(p) 75.3	(q) N/A
(4)	(5)		(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	-13.8	-9.8	-9.3	6.0	24.7	36.1	42.3	42.0	36.2	24.4	4.7	-6.1
	DBStd	23.16	12.85	15.34	13.02	10.96	8.88	4.78	4.78	5.12	5.25	6.33	12.08	13.09
	HDD50	12823	1978	1675	1837	1319	786	419	246	257	413	794	1359	1740
	HDD65	18281	2443	2095	2302	1769	1251	867	705	713	863	1259	1809	2205
	CDD50	19	0	0	0	0	0	2	6	10	1	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	14.4	13.9	14.9	14.1	13.2	13.6	14.9	14.0	14.8	16.5	16.6	14.1	12.8
	PrecAvg	7.3	0.4	0.3	0.3	0.2	0.3	0.6	1.1	1.5	1.1	0.9	0.5	0.4
	PrecMax	13.2	1.2	0.9	1.3	0.5	0.9	1.7	2.6	3.5	3.0	2.0	1.3	1.3
(15) Precipitation	PrecMin	4.3	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.4	0.2	0.4	0.1	0.0
	PrecStd	1.9	0.3	0.2	0.2	0.1	0.2	0.4	0.5	0.8	0.7	0.4	0.3	0.3
	DB	27.3	29.6	28.4	36.9	42.0	60.4	61.1	61.9	55.0	39.6	31.3	27.2	
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	11.7	23.6	20.7	30.9	38.0	52.9	56.1	56.7	50.3	35.9	25.3	21.8	
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% DB	6.5	16.8	13.8	25.6	36.3	46.7	52.7	53.1	46.6	33.1	23.0	17.1	
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10% DB	3.0	11.5	7.7	21.4	34.7	41.8	49.0	49.8	42.6	31.7	20.3	11.6	
	MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	MDBR	10.6	12.7	13.1	12.8	8.7	7.7	7.7	6.4	4.9	6.3	10.3	10.5	
	5% DB	15.5	19.2	14.4	14.6	9.5	16.5	14.2	11.9	12.2	5.3	10.8	13.9	
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(40) Clear-Sky Solar Irradiance	taub	N/A	0.198	0.239	0.296	0.317	0.330	0.344	0.330	0.295	0.217	N/A	N/A	
	taud	N/A	2.046	2.204	2.114	2.160	2.297	2.411	2.517	2.533	2.171	N/A	N/A	
	Ebn at Noon	0	171	249	262	271	272	264	255	233	159	0	0	
	Edh at Noon	0	11	22	34	37	34	29	23	17	10	0	0	
(44) All-Sky Solar Radiation	RadAvg	0	107	610	1351	1872	1996	1628	932	462	133	0	0	
	RadStd	0	9	25	63	92	121	120	93	42	18	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+2.05	N/A	N/A	N/A	N/A	N/A	-902	-889	N/A	N/A

Nomenclature: See separate page